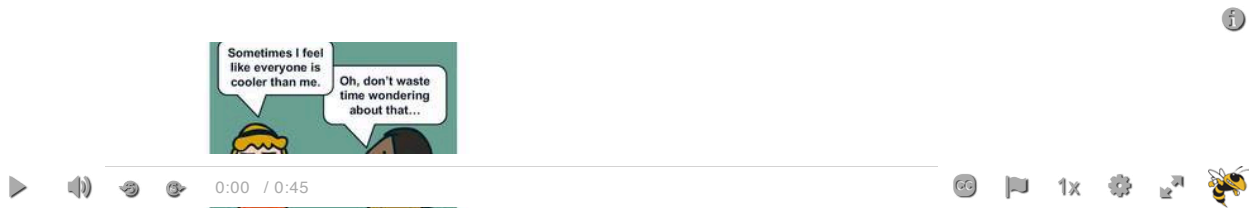


L3: Friendship Paradox



The Probability that a Random Edge Connects to a Node of Degree k

Let's start by deriving a simple fact that we will use repeatedly in this course.

Suppose that we pick a random edge in the network – and we randomly select one of the two end-points of that edge -- we refer to those end-points as the **stubs** of the edge. What is the probability q_k that a randomly chosen stub belongs to a node of degree k ?

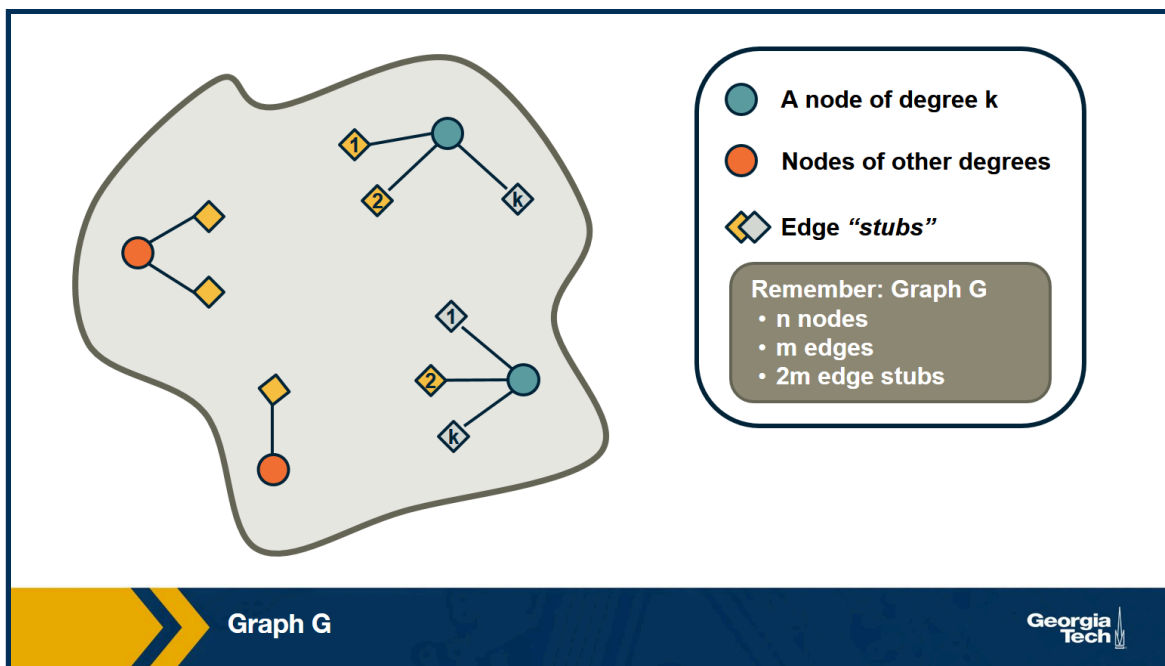
This is easy to answer when the degrees of connected nodes are independent.

$q_k = (\text{number of nodes of degree } k) \times (\text{probability an edge connects to a specific node of degree } k)$

$$= (np_k) \left(\frac{k}{2m} \right) = \frac{kp_k}{\frac{2m}{n}} = \frac{kp_k}{\bar{k}}$$

Note that the probability that the randomly chosen stub connects to a node of degree k is proportional to both k and the probability that a node has degree k .

This means that, for nodes with degree $k > \bar{k}$, it is more likely to sample one of their stubs than the nodes themselves. The opposite is true for nodes with degree $k < \bar{k}$.



Based on the previous derivation, we can now ask: *what is the expected value of a neighbor's degree?*

Note that we are not asking for the average degree of a node. Instead, we are interested in the average degree of a node's neighbor.

This is the same as the expected value of the degree of the node we select if we sample a random edge stub. Lets denote this expected value as $\overline{k_{nn}}$.

The derivation is as follows.

$$\overline{k_{nn}} = \sum_{k=0}^{\max \text{ degree}} k \cdot q_k = \sum_k k \frac{k p_k}{\bar{k}} = \frac{\sum_k k^2 p_k}{\bar{k}} = \frac{\overline{k^2}}{\bar{k}} = \frac{(\bar{k})^2 + (\sigma_k)^2}{\bar{k}} = \bar{k} + \frac{\sigma_k^2}{\bar{k}}$$

We can now give a mathematical statement of the friendship paradox: as long as the variance of the degree distribution is not zero, and given our assumption that neighboring nodes have independent degrees, the average neighbor's degree is higher than the average node degree.

The difference between the two expected values (i.e., $\sigma_k^2 / \bar{k}$) increases with the the variability of the degree distribution.

Food for Thought

Can you explain in an intuitive way why the average neighbor's degree is larger than the average node degree, *as long as the degree variance is not zero?*